# pyAmpliCol at one loop

Batched Born–loop interference · implementation & live benchmarks

2026-10-18 · measured revision fa599885 · AMD EPYC 9754

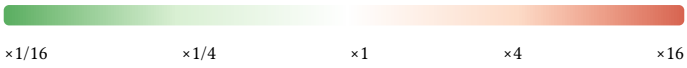
**Reading guide.** Each coloured cell in the summary table shows **pyAmpliCol time**, the **fastest available reference** for the same process and batch, and their ratio. References are MadLoop (ML), Recola and OpenLoops (OL); the winner is selected independently in each cell. Green is faster, white parity, red slower. These are complete Born–loop results: all helicities, full SU(3), UV, model-rule $R_2$ and all three Laurent coefficients.

## Complete evaluation, from cold plan to warm batch

Warm values are **µs per point**, medians of five calls on a repeated three-point stream. Every call recomputes events. Generation is in seconds, with first-call latency and peak RSS below it; no provider generation ratio is available.

Table 1: **Complete binary64 controls.** The named reference and runtime multiplier appear below each value.

| Process | Generation s | $B = 1$ µs / point | $B = 8$ µs / point | $B = 32$ µs / point | $B = 128$ µs / point |
|---|---|---|---|---|---|
| $u\bar{u} \to e^-e^+$<br>1 loop · 1 family | 2.43 s<br>first 21.6 ms<br>RSS 132 MiB | 583.7<br>ML 46.86 µs<br>×12.46 | 170.0<br>ML 46.37 µs<br>×3.67 | 142.0<br>ML 46.84 µs<br>×3.03 | 147.4<br>ML 46.49 µs<br>×3.17 |
| $u\bar{u} \to d\bar{d}$<br>14 loops · 8 families | 7.09 s<br>first 63.0 ms<br>RSS 180 MiB | 1733<br>ML 251.3 µs<br>×6.90 | 814.9<br>ML 254.7 µs<br>×3.20 | 719.4<br>ML 255.6 µs<br>×2.81 | 764.0<br>ML 254.3 µs<br>×3.00 |
| $u\bar{u} \to t\bar{t}$<br>14 loops · 9 families | 6.39 s<br>first 90.6 ms<br>RSS 183 MiB | 2006<br>ML 352.0 µs<br>×5.70 | 940.0<br>ML 335.2 µs<br>×2.80 | 811.9<br>ML 318.7 µs<br>×2.55 | 745.0<br>ML 376.2 µs<br>×1.98 |
| $u\bar{u} \to gg$<br>45 loops · 20 families | 21.16 s<br>first 616.2 ms<br>RSS 321 MiB | 6835<br>ML 462.8 µs<br>×14.77 | 4702<br>ML 407.6 µs<br>×11.54 | 4125<br>ML 404.4 µs<br>×10.20 | 3100<br>ML 371.5 µs<br>×8.34 |
| $gg \to gg$<br>186 loops · 52 families | 98.74 s<br>first 11531.7 ms<br>RSS 1085 MiB | 33 656<br>ML 2593 µs<br>×12.98 | 19 120<br>ML 2327 µs<br>×8.22 | 15 208<br>ML 2283 µs<br>×6.66 | 14 280<br>ML 2073 µs<br>×6.89 |




Logarithmic scale · lower is faster · white is parity · red is slower

### Cold and first-call costs

Generation includes UFO capture, graphs, colour, recurrence, CT matching and JIT/reducer plans. Imports and installation are excluded. First evaluation is $B = 3$, timed separately. RSS is the high-water mark of the full run, not an event-buffer estimate.

### Comparable physics; different boundaries

Coefficients and points agree, but pyAmpliCol includes exact decimal ingress and its Python batch coordinator. Provider timers start with native arrays. Provider rescue policies remain active; pyAmpliCol has no rescue in these measurements. Output formatting is excluded.

**Development comparison, not acceptance.** Complete totals are slower than MadLoop in every listed case. Required high-multiplicity gates remain: faster than MadLoop and at most 1.30 × each available Recola/OpenLoops result, for latency and native throughput.

CPU 382, one thread, shared host. Rust dev: local opt3 / dependencies opt2, debug/overflow checks retained, no LTO; Community opt2 and SymJIT. Five samples describe local variation. EW, direct-$R_2$ and higher multiplicity are not timed here.

MEASUREMENTS · COMPONENTS AND REFERENCES

# The work behind a complete event

The native component clocks separate numerator execution, integrand reduction and scalar masters. Everything else remains visible. These boundaries differ from MadLoop's, so stage cells are deliberately neutral: **no stage speed ratio is inferred**.

Table 2: **pyAmpliCol at $B = 32$.** µs per point; medians of five instrumented calls. Independently computed medians need not add to the median total.

| Process | Numerator | NinjaRS reduction | Scalar masters | Other work | Instrumented total |
|---|---|---|---|---|---|
| $u\bar{u} \to e^-e^+$ | 24.67<br>DAG + callbacks | 0.92<br>exclusive engine | 0.35<br>oneloopmaster | 71.69<br>states · UV / $R_2$ | 96.84<br>with clocks |
| $u\bar{u} \to d\bar{d}$ | 294.0<br>DAG + callbacks | 7.76<br>exclusive engine | 9.05<br>oneloopmaster | 381.0<br>states · UV / $R_2$ | 691.4<br>with clocks |
| $u\bar{u} \to t\bar{t}$ | 334.9<br>DAG + callbacks | 16.10<br>exclusive engine | 18.28<br>oneloopmaster | 520.8<br>states · UV / $R_2$ | 889.8<br>with clocks |
| $u\bar{u} \to gg$ | 1778<br>DAG + callbacks | 21.52<br>exclusive engine | 17.65<br>oneloopmaster | 1389<br>states · UV / $R_2$ | 3196<br>with clocks |
| $gg \to gg$ | 9185<br>DAG + callbacks | 88.42<br>exclusive engine | 68.76<br>oneloopmaster | 4515<br>states · UV / $R_2$ | 14 167<br>with clocks |

Table 3: **Complete provider calls at $B = 32$, µs per point.** Each total is coloured against the fastest existing reference. ML's smaller lines give its separately instrumented numerator and reduction + masters; these are not components of the uninstrumented total above them.

| Process | MadLoop | Recola | OpenLoops | pyAmpliCol |
|---|---|---|---|---|
| $u\bar{u} \to e^-e^+$ | 46.84<br>×1.00<br>num 6.47 · red+M 35.86 | —<br>not measured | —<br>not measured | 142.0<br>ML 46.84 µs<br>×3.03 |
| $u\bar{u} \to d\bar{d}$ | 255.6<br>×1.00<br>num 55.76 · red+M 197.1 | 2294<br>ML 255.6 µs<br>×8.98 | 543.0<br>ML 255.6 µs<br>×2.12 | 719.4<br>ML 255.6 µs<br>×2.81 |
| $u\bar{u} \to t\bar{t}$ | 318.7<br>×1.00<br>num 94.48 · red+M 256.9 | 2953<br>ML 318.7 µs<br>×9.27 | 675.9<br>ML 318.7 µs<br>×2.12 | 811.9<br>ML 318.7 µs<br>×2.55 |
| $u\bar{u} \to gg$ | 404.4<br>×1.00<br>num 205.5 · red+M 127.3 | 4884<br>ML 404.4 µs<br>×12.08 | 1485<br>ML 404.4 µs<br>×3.67 | 4125<br>ML 404.4 µs<br>×10.20 |
| $gg \to gg$ | 2283<br>×1.00<br>num 1916 · red+M 824.8 | 32 811<br>ML 2283 µs<br>×14.37 | 10 358<br>ML 2283 µs<br>×4.54 | 15 208<br>ML 2283 µs<br>×6.66 |

### pyAmpliCol clocks

Numerator includes loop DAGs, reducer callbacks and coefficient-head preparation. Reduction excludes callbacks and masters. Other work contains states, exact kinematics, Born and UV/$R_2$ currents, weighting, assembly, transport and clock overhead. Master timing retains the adapter's cache policy.

### Reference clocks

ML numerator excludes shared HELAS wavefunctions; its loop-CT window includes reduction interfaces, masters and retries. Recola/OL totals use native public calls in compiled loops. Original-binary totals and instrumented stages are separate observations, with all raw samples retained.

**Next benchmark gate.** Use distinct frozen event streams, paired timing blocks, matched stability criteria and native-interface boundaries; report uncertainty, rescue rates and memory. A fast average cannot hide a failing designated process. The current three-point developmental stream does not establish those acceptance conditions.

ARCHITECTURE · PHYSICAL CONTRACT

# The quantity being computed

The new engine targets one-loop amplitudes interfered with a Born amplitude. It does not compute the squared one-loop amplitude and does not admit loop-induced production. A Born zero at an individual point or helicity remains a valid event of a supported process.

For Born order $a$, loop order $b$ and squared coupling order $s = a + b$, the ordinary uncorrelated observable is

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\Re\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0, -1, -2\}.$$

The rectangular matrix $G_{BL}$ connects the actual Born and loop colour bases. The normalization $\mathcal{N}$ includes declared initial averages and final identical-particle factors. Couplings are included; the result is neither divided by the Born value nor normalized by an extracted strong coupling.

The returned coefficient order is **(finite, single pole, double pole)** in the MadLoop convention,

$$V = C_\varepsilon \left( V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2} \right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}.$$

The finite conversion used in the matched Recola controls is $V_0^{\mathrm{ML}} = V_0^{\mathrm{Recola}} + \left(\frac{\pi^2}{6}\right) V_{-2}$; both poles are unchanged. Such conversions act on the Laurent series, not on a process-dependent fitted factor. Scheme, scale and finite renormalization conventions are retained separately.

## Coupling selection is a graph property

`loop_coupling_orders` requests perturbative corrections. Eligibility depends on the actual loop interactions and internal species, followed by explicit Born–loop pairings. An order difference alone cannot identify the correct graphs. In the mixed four-quark control,

$$B = B_{20} + B_{02}, \quad L = L_{40} + L_{22} + L_{04},$$

the squared order $(4, 2)$ receives both $B_{20}^* L_{22}$ and $B_{02}^* L_{40}$. Their identities survive through counterterm matching and output assembly. Joint QCD/QED selection also admits mixed graphs absent from either singleton selection. Independent MadLoop graph controls give 16 QCD, 15 QED and 35 joint loops for the retained mixed model.

## What the measured processes include

| Control | Initial / identical normalization | Physical completion |
|---|---|---|
| $u\bar{u} \to e^- e^+$ | $\frac{1}{36}$ | Photon Born; QCD virtual |
| $u\bar{u} \to d\bar{d}$ | $\frac{1}{36}$ | Local UV and $R_2$ |
| $u\bar{u} \to t\bar{t}$ | $\frac{1}{36}$ | Local and external-field UV; $R_2$ |
| $u\bar{u} \to gg$ | $\frac{1}{72}$ | Local/field UV; $R_2$ |
| $gg \to gg$ | $\frac{1}{512}$ | Local/field UV; $R_2$ |

The QCD scattering controls use $\alpha_s = 0.118$, $\mu_R = 1000$ GeV, stable $m_t = 173$ GeV, five renormalization flavours and six quarks in loops; the other quark masses vanish under explicit control restrictions. Drell–Yan uses $\alpha^{-1} = 132.507$. Initial spin and colour are averaged, and physical external helicities are summed. The two final-gluon controls include $\frac{1}{2}$.

ARCHITECTURE · MODEL AND COUNTERTERMS

# Preserve the model before lowering it

Loop models contain information that a tree-only loader can discard: Laurent-valued parameters and couplings, UV and $R_2$ categories, loop-particle groups, external-field factors and duplicate vertex names. The new raw model retains positional declaration identities and these fields before constructing the exact scalar and covariant-kernel representations.

**Raw source**
Exact declarations
Species and loop groups

→

**Prepared model**
Scalar programs
Colour × Lorentz entries

→

**Process plan**
Admitted graphs
Matched UV / $R_2$ / fields

## Exactness begins at input

Statically identifiable source literals are preserved as text, including long decimal and Laurent coefficients. String-valued expressions remain exact. Arbitrary Python computations or mutations do not gain an exactness claim merely because their imported result is a number. Explicit exact runtime overrides and fixed parameters flow through the same retained scalar owner.

Symbolica performs substitution, series extraction and expression preparation. Source helper functions and their branch semantics are retained. Each selected scalar has a value and an active diagnostic expression; native evaluation checks both. A nonfinite or unsupported active branch cannot silently inherit a successful amplitude status. Per-point renormalization scales reach the same scalar program used by counterterms.

## Tensor structure and physical frame

Each basis entry retains its scalar coefficient, native Lorentz tensor, native colour tensor and ordered particle ports. Multi-colour quartic vertices remain sums of local terms. There is no reconstruction of tensor algebra in pyAmpliCol: Idenso/Spenso/FeynKit own contractions, traces, component libraries and index transformations.

Source spinor slots and the native physical wavefunction convention require an explicit, bounded frame map. The FFV map renames typed external Dirac ports simultaneously. The newly admitted bounded FFS map also exchanges `ProjP`/`ProjM`, with 580 independent ALOHA controls; the FFV map and propagators remain unchanged. Neither map rewrites momentum labels or supplies a universal gamma-five transpose. Unvalidated source structures retain an explicit boundary.

## Counterterms follow admitted loops

UV-tree terms, local loop counterterms and external-field factors have different eligibility rules. A standalone counterterm tree is insufficient evidence that its loop correction belongs to the request. Local $R_2$/UV matching retains the actual attachment currents, ordered binding legs, local coupling order and allowed loop-species alternatives. Wildcards and repeated alternatives preserve their source semantics and multiplicity. Fresh native tree graphs provide the insertion signs and propagators.

External-field factors multiply the applicable Born currents with each physical leg retained. Local and field UV remain separately inspectable, then join the same ordered interference. Model-rule $R_2$ currents already contain their loop normalization; the assembly does not multiply them by a second loop measure.

The bounded Gmu continuation offers an explicit `ufo-standard` logarithm convention. The default preserves the source helper; either choice retains the original model and does not certify complete CMS physics.

**Evidence boundary.** Complete bounded QCD controls and real-mass neutral-current charged-lepton EW components are validated. That does not certify every rule in a supplied UFO, arbitrary restriction cards, all gauges, or general BSM renormalization.

ARCHITECTURE · RECURRENCE AND FAMILIES

# Local tensors, reusable coefficient currents

The production direction is a dedicated loop recurrence. FeynKit supplies the graphs, directed incidence, loop signs, symmetry factors, cuts and affine momentum routing. pyAmpliCol turns those local ingredients into a coefficient dataflow graph and schedules its numerical evaluation.

A current retains the coefficients of a symmetric polynomial in the physical loop momentum. For a local tensor $T$ and previous current $J$, the next step is the contraction $T \cdot J$, followed by coefficient extraction using Symbolica. Only the local transition is prepared symbolically. A whole-diagram symbolic trace is used in small independent oracles, not as the numerical recurrence.

**Coefficient dataflow.** External states and tree attachments feed local transitions; loop closure produces ordered coefficient outputs. Native saved evaluator instructions update these outputs in batches. Each original graph retains its colour terms, coupling sector and routing map even when it shares an identical numerical schedule with another graph.

## Topology remains attached to the numerator

Reduction needs a denominator set as well as a polynomial. Each graph carries its routed propagators and a dependency-proved map into an integral family. Numerators are transported through that map before combination. Cyclic or reversed order is not inferred from a loose list of masses: momenta, incidence, external attachments and the actual transformation must agree.

The family plan retains bridges and local tree denominators. Native graph factors carry fermion and ghost signs. Multi-term local vertices are expanded into ordered recurrence outputs, while the original graph/family identities remain intact. This separates a numerical reuse opportunity from a physical multiplicity or symmetry factor.

## Contract before reducing when linearity permits

For an interference observable, Born amplitudes and the rectangular colour weights can multiply numerator coefficients before integration. Compatible graphs, helicities and colour terms then share a reduction within their family and coupling sector. The reducer sees the coefficient combination needed by the observable, rather than every independent loop amplitude.

Complex weights are retained until the correctly directed bilinear is formed. Taking a real part too early loses information needed by colour histories and other off-diagonal observables. Counterterm interference uses the same Born currents and colour conventions, with its own local scalar factors.

## Sharing that is safe under parameter updates

Compiled kernel reuse keys the saved instructions, settings, carrier precision and output layout. Evaluators may share immutable compiled code while owning independent work buffers. Whole-schedule numerical reuse additionally requires identical ordered inputs under an exact model-binding proof. Equal floating values at one point are insufficient.

The current process cache shares compatible Born, loop and counterterm kernels; exact whole-schedule groups cover Born and loop execution. It does not cache event results between calls. Complete unprojected model-rule QCD plans now retain exact instructions and bindings across arithmetic views. Rebinding creates fresh evaluators, reducers and buffers without repeating graph, colour or coefficient preparation. Colour-weight changes invalidate the cached plan. Source-model, direct-$R_2$ and formal-colour reuse remain outside this checkpoint; persistent artifacts and broader subtree reuse are still open.

The staged NinjaRS path supports up to six propagators and validated weighted rank through $n + 1$; unsupported ranks reject. The owner's generic-any-$n$ extension (`e4c16a7`) has been pulled. A separate consumer candidate is testing seven propagators in all three carriers; staged-module adoption and required high-multiplicity physics remain pending. No timing table uses that candidate.

ARCHITECTURE · COLOUR AND OBSERVABLES

# Full colour without dense event tensors

The current complete QCD path contracts colour at plan construction using native fixed-SU(3) tensors. It forms the properly scoped bra–ket expression, uses dependency colour simplification and caches the exact pair. The numerical event path receives the resulting rectangular weights. Dense SU(3) components remain an independent small-control oracle, not the production colour sum.

Closed residual contractions use the dependency's existing projector expansion, trace/chain operations and canonicalization. Minimal upstream fixes address specific demonstrated tensor-lowering or invariant-contraction defects. No private group identities or numerical colour fitting are inserted into the application.

## LC and NLC are structural projections

For a certified formal-$N_c$ model, all contributions belonging to one squared order and operator family are combined before finding the common highest power $P$. With $N_f$ held fixed,

$$\mathrm{LC} : N_c^P, \quad \mathrm{NLC} : N_c^P, N_c^{P-1}, N_c^{P-2}.$$

Odd suppression powers are retained. The same $P$ spans the relevant helicities, epsilon sectors and source contributions; choosing a different leading power for each term would change the approximation. Exact native algebra handles Laurent coefficients and semantic $C_A, C_F, T_R$ relations.

Complete bounded quark-process colour projections have controls. The active `loop_sm` four-gluon $R_2$ model contains fixed-$N_c = 3$ coefficients. Its full-SU(3) validity does not imply a formal continuation, and numerical constants are never promoted heuristically to $N_c$. Validated direct completion can provide a different continuation only within its certified domain.

## Directed colour and spin sources

Correlated virtual observables require both rectangular cross terms. For operator histories $A$ and $B$, their generic form is

$$B_A^\dagger G L_B + L_A^\dagger G B_B.$$

For unequal histories this is not generally $2\mathfrak{R}\left(B_A^\dagger G L_B\right)$; the result can be complex. Existing colour-connection metadata and native charge tensors provide the operator action and signs. Current internal controls compose bounded histories with Born, bare loop, local counterterms and field factors.

Literal complex spin-source vectors enter the admitted physical external ports without an implicit conjugation, normalization or projection. Spectator states are summed and a replaced leg is counted once. The source/current frame and crossing factors remain explicit. Scalar and massive-vector layouts reuse native wavefunctions; a generic source API is not yet certified by these controls.

## Flavour mapping and FFT remain distinct work

An exact species-symmetry certificate now compares the entire retained ordinary, UV, $R_2$ and external-field plan, including prepared native structures and parameter bindings. Independent massless $d$ and $s$ catalogues pass; unequal masses, Yukawas or UV logarithms reject. This is a prerequisite for mapping, not yet external-leg, momentum, spin-source or colour-history transport.

Optional FFT contraction is an agreed feature but is not part of the measured path. Its cost model, exact observable reconstruction and composition with the other features remain acceptance gates.

ARCHITECTURE · NATIVE BATCHES AND PRECISION

# One numerical path, three real carriers

The internal numerical core uses binary64, DoubleFloat and arbitrary-precision carriers throughout coefficient evaluation, kinematics, NinjaRS reduction, scalar masters, counterterms and contractions. It does not compute a binary64 answer and promote it afterward. Fixed-carrier precision metadata follows the actual numerical types; arbitrary precision carries its requested precision.

Exact decimal or rational input is parsed before carrier conversion. S-matrix and admitted on-shell invariants are constructed through the exact input boundary; source mass/coupling formulas are evaluated natively in the chosen carrier. Controls with differences below binary64 resolution verify that DoubleFloat and arbitrary precision retain the input information through the output. Source scalar callbacks and their diagnostics use those same carriers.

## Batching is an execution contract

Points, helicities and coefficient components have explicit ordered layouts. Native buffers retain original point identities through compaction, tail batches, failure masks and result scatter. Changing sizes, empty batches and repeated calls are covered. Python coordinates batch plans; it does not evaluate each phase-space point in an interpreted numerical loop.

The coefficient scheduler uses bounded row tiles and reusable scratch storage. Its present 16-row default is a measured development choice, not a universal optimum. Dependency evaluators own their arithmetic workspaces. Binary64 can use SymJIT; high-precision paths remain native batches without an unsupported SIMD claim. Public standalone SDK integration is a separate unfinished layer.

## Reducer and master interfaces

NinjaRS receives the polynomial layout, routed denominator data, exact kinematic invariants and carrier-appropriate inputs. The adapter retains weighted rank: $\mu^2$ counts as two powers of loop momentum. Separate explicit-epsilon sectors reduce independently and combine by Laurent multiplication. A newer dense-$\mu$ numerator adapter is available as an additive internal option; the headline timings specify the actual `fast` engine they use.

The native module and Community producer use the same underlying numerical and symbolic owners. Python distribution and Rust kernel version labels are recorded separately. A genuine isolated development-package install now passes a public ordinary-tree generation/evaluation smoke. Publishable dependency packaging, loop artifacts and cross-language virtual SDK delivery remain open.

## Success, accuracy and rescue are different facts

A reducer's success flag is not an accuracy certificate. Native checks reject nonfinite input, state, coefficient and counterterm values, preserve the original failed point identity, and permit an explicit partial-result contract. Optional cross-precision checks compare complete outputs and can rescue selected failed or disagreeing points at higher precision without changing $R_2$ strategy.

The result records requested/evaluated precision, strategy, status and available agreement evidence. Failed points cannot silently appear as zero or be declared accurate. Existing bounded rescue controls are not a universal stability policy: threshold choices, complete feature composition and adversarial tests remain open. None of the headline runtime cells enables rescue or cross-precision verification, and each is labelled accordingly.

**Known numerical gate.** An equal-value, differently tracked arbitrary-precision mass construction exposed a scalar-box branch discrepancy in a CMS zero-width control. This is retained as a dependency-owner accuracy issue. Extra digits alone are not evidence that a branch-sensitive result is correct.

ARCHITECTURE · RATIONAL TERMS

# Two complete strategies, one observable

The model strategy combines the four-dimensional numerator contribution with the model's effective $R_2$ rules and UV terms. The direct strategy instead computes the required evanescent numerator remainder and retains the same UV completion. The strategy is explicit in the result; a caller cannot relabel an executed model result as direct.

Four-dimensional reduction already produces denominator-related rational terms. The presence of $\mu^2$ in a reducer interface therefore does not, on its own, prove that numerator-origin $R_2$ is complete. The direct extension must account for extra-dimensional momentum, metric/spin traces and explicit epsilon factors.

## Local evanescent state, not a whole-loop trace engine

For a certified closed vectorlike fermion cycle, the local component construction uses dependency gamma matrices with one auxiliary evanescent slash. Odd auxiliary powers survive every open transition; only the closed trace selects even parity. Independent full-$D$ local-graph oracles check routed bubbles, triangles and boxes. The auxiliary gamma-five realization is not a general physical chiral or dimensional gamma-five prescription.

For the admitted massless Yang–Mills cycles, dependency metric and momentum rules are continued to an explicit physical/aligned/spectator state space. The remaining transverse-state multiplicity is retained symbolically in $D = 4 - 2\varepsilon$. Local cubic and quartic colour terms, closure, routing and spectator weighting are checked independently. Ghost cycles keep their native statistics and their ordinary bare contribution.

## Epsilon sectors matter after integration

If the numerator is $N_0 + \varepsilon N_1 + \varepsilon^2 N_2$ and each reduced sector has finite/single/double coefficients $\left(F_j, S_j, D_j\right)$, the assembled result is

$$V_0 = F_0 + S_1 + D_2, \quad V_{-1} = S_0 + D_1, \quad V_{-2} = D_0.$$

Dropping explicit epsilon terms can therefore change the finite answer even when the mu-dependent terms are retained. Native reduction controls exercise this convolution in all three carriers. Source templates, full-graph algebra and integrated physical results have separate checks.

## What is supported internally today

The complete retained full-colour $gg \to gg$ control admits the local direct kernels, includes UV completion, and agrees with model-rule components and independent reference results in all three arithmetic modes. Bounded cross-precision verification and direct-strategy rescue are also implemented. Public direct dispatch remains behind its integration gates.

Open fermion chains, general mixed-spin loops, massive gauge/Goldstone sectors, physical axial or anomaly-sensitive structures and arbitrary BSM dimensional extensions are not inferred from that success. Unsupported direct domains need compatible model-rule completion. No universal massless FDH-to-HV shift is assumed for massive EW or general models.

**Selection and performance.** Automatic policy may prefer a certified direct strategy, but completeness is mandatory. Model and direct costs must be measured as standalone complete evaluations. The present headline table uses model rules throughout; a new direct-versus-model comparison is still due after recent colour and scheduling changes.

VALIDATION · NEXT GATES

# What this snapshot establishes

The measured controls are useful working vertical slices: model import, graphs, local recurrence, full colour, native states, reduction, UV/$R_2$ and ordered physical output all run together. Independent reference comparisons separate bare, local UV, external-field UV and $R_2$ so cancellations in the complete answer cannot conceal a wrong component.

| Area | Current boundary |
|---|---|
| Complete QCD | Internal DY, four-quark, top-pair, quark/gluon and gluon controls; all three carriers. |
| Real-mass EW | Source-owned neutral-current charged-lepton controls, including bounded massive muons; complete components and scale/parameter changes. |
| CMS | Explicit Gmu log choice matches MadLoop components: double/DD across widths, AP80 at finite/small widths. The AP80 zero-width master gate and Recola finite-UV convention remain separate. |
| Correlations / mapping | Bounded directed/spin controls and exact species symmetry; general composition and mapping transport remain open. |
| Public delivery | A genuine development install passes the public tree smoke; loop artifacts, commands and SDK/UMAMI integration remain open. |
| Performance | Current development timings only; distinct-point matched provider gates and required high multiplicity remain open. |

## Required multiplicity and acceptance

The target includes selected $2 \to 5$ processes, with $2 \to 6$ as a stretch goal. The common-provider set includes gluon jets, heavy-quark plus jets, lepton-pair plus jets, and heavy-quark/lepton-pair production with a jet. Five-final-parton gluon channels also require MadLoop/Recola controls; an unavailable public OpenLoops channel is not invented for comparison.

The new generic-any-$n$ reducer still needs consumer validation and adoption; seven-propagator physics is a separate gate. Rank above the supported $n + 1$ domain must reject. Required progress also includes broader cold-plan coverage, persistent artifacts, optional FFT, complete correlation/mapping transport and broader certified model coverage. These requirements remain visible even when a smaller control is fast.

## Independent evidence and reproducibility

The accompanying `data/measurements.json` contains the actual five timing samples, revision, process counts, cold times and checksums behind the headline tables. The frozen point fixtures and native reference drivers live in `tools/loop_reference/`. The measured driver calls the complete internal `evaluate_ordered_native` path; checks and decimal result conversion occur after the event timer. Test durations and earlier bare-only profiles are not event timings in this report.

Provider controls use MadLoop 3.7.0, Recola2 2.3.0 with its matched SM/Collier model, and OpenLoops 2.1.5. Their independent component tests establish the bounded normalization comparisons; only the fresh matched-host control timings enter the current performance table, with their input and rescue asymmetries declared. This report follows the visual structure of the NinjaRS benchmark while measuring a different object: a complete physical interference, not a single diagram and helicity.

Further reading: repository `docs/development/LOOP_MODEL_IR.md`, `docs/development/LOOP_DIRECT_R2.md`, `docs/loop-flavor-symmetry.md`, and `tools/loop_reference/README.md`; OpenLoops on-the-fly reduction · Recola · four-dimensional formulation · MadLoop standalone convention.

**Living document.** Update the source, measured data and compiled PDF together after each implementation milestone. Keep incomplete rows and capability limits explicit. The completion criterion is the full physics, feature, precision, packaging and per-process performance contract—not the existence of this first report.